
Intrinsic Token Weighting for Credit Assignment in Language Model Reinforcement Learning

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Abstract

1 Policy gradient methods for LLM reinforcement learning often rely on value-
2 function critics, but language-generation prefixes are non-Markov and poorly
3 suited for stable value estimation. Recent critic-free approaches such as GRPO
4 and RLOO avoid value networks, yet they still broadcast the same trajectory-level
5 signal uniformly across tokens. We study an alternative: intrinsic token weighting,
6 which redistributes a prompt-level advantage across tokens using salience signals
7 already available from the policy, including token surprisal and local entropy
8 reduction. These methods require no additional networks, no tree construction,
9 and no reward models. To make the experimental setup concrete and reproducible,
10 the repository implements Qwen-based experiments on GSM8K and MBPP with
11 exact-answer and unit-test rewards. The resulting codebase provides a direct way
12 to evaluate whether redistributing credit via intrinsic distributional signals is a
13 practical alternative to uniform token updates in critic-free language RL.

14 1 Introduction

15 Reinforcement learning has become an important component of large language model (LLM) training
16 pipelines, particularly for alignment, preference optimization, and reasoning tasks. Most contem-
17 porary approaches rely on policy gradient methods augmented with advantage estimation, typically
18 implemented via learned value functions or generalized advantage estimation. These techniques orig-
19 inate from classical reinforcement learning settings in which states are well-defined, approximately
20 Markov, and admit stable value estimates.

21 Applying this framework to language generation introduces a fundamental mismatch. In autore-
22 gressive LLMs, the “state” is a partial sequence of tokens. Such prefixes are not Markov, do not
23 summarize progress toward task completion, and can vary widely in semantic meaning despite
24 superficial similarity. Consequently, estimating a value function over prefixes is both statistically
25 challenging and conceptually ill-defined. Empirically, value heads in language RL are often brittle,
26 sensitive to distributional shift, and prone to introducing bias that outweighs their variance-reduction
27 benefits.

28 Motivated by these issues, recent work has increasingly moved away from learned critics in LLM
29 reinforcement learning. Critic-free, outcome-based methods such as RLOO and GRPO replace value
30 estimation with batch-level baselines, achieving stable training on verifiable reasoning tasks such as
31 math and code. However, these approaches still implicitly assume uniform credit assignment across
32 tokens: every action in a trajectory receives the same reward signal, modulated only by a scalar
33 baseline.

34 This raises a natural question: *if we no longer estimate values over prefixes, how should trajectory-*
35 *level reward be distributed across tokens?* Uniform credit assignment is simple, but it ignores

the fact that different tokens play very different roles in generation. Some tokens correspond to high-uncertainty decisions, strategy commitments, or branching points in reasoning, while others are low-entropy continuations or stylistic filler. Treating all tokens equally risks diluting the learning signal and obscuring the mechanisms by which policies improve.

In this work, we study token-level credit redistribution as a mechanism for on-policy LLM reinforcement learning without value functions. Rather than estimating advantages or prefix values, we propose to weight token-level policy gradients using intrinsic signals derived directly from the model’s predictive distribution. These include token surprisal and changes in predictive entropy, which naturally reflect decision difficulty and commitment during generation. Importantly, these signals are inexpensive to compute, require no additional networks or tree structures, and make no assumptions about the existence of a meaningful value function over prefixes.

To make this question concrete, the repository now implements a reproducible experimental stack around Qwen models on GSM8K and MBPP. The code compares uniform token updates against RLOO/GRPO baselines with surprisal-based and entropy-reduction-based intrinsic weighting, all under matched prompt formatting and optimization settings. This turns the paper’s experimental section into a directly runnable benchmark rather than a placeholder description.

Overall, this paper reframes advantage estimation as one instantiation of a broader credit assignment problem in language reinforcement learning. In settings where state semantics are weak and rewards are trajectory-level, redistributing credit using intrinsic distributional signals is a plausible alternative to value-based methods; the implemented experiments are designed to test that claim directly.

2 Related Work

2.1 From RLHF to RLVR

Modern language-model post-training inherits its optimization machinery from policy-gradient reinforcement learning, from REINFORCE [25] to trust-region and clipped-surrogate methods such as TRPO and PPO [16, 18]. Advantage estimation, especially GAE, became the standard variance-reduction device in continuous-control RL by learning value functions over states or prefixes [17]. In language modeling, these ideas entered mainstream use through RLHF systems that optimize sequence-level rewards derived from human preferences or reward models [5, 21, 14].

The recent reasoning-model wave shifted part of the field from preference-based RLHF toward reinforcement learning with verifiable rewards (RLVR), where supervision is often a deterministic correctness signal on math or code tasks. DeepSeekMath introduced GRPO as a practical critic-free alternative for these settings [19], and DeepSeek-R1 popularized large-scale RL-only or RL-dominant reasoning pipelines [7]. Subsequent surveys document how quickly this regime became the default template for open reasoning-model replication and extension [29]. This transition matters for our setting because RLVR makes sparse outcome rewards and long reasoning trajectories central, thereby exposing the credit-assignment problem more directly than earlier short-form RLHF tasks.

2.2 Critic-Based and Critic-Free Policy Optimization

The classical solution to delayed reward is to learn a critic or value function and convert returns into token- or prefix-level advantages. PPO-style RLHF pipelines follow this recipe, but the value-estimation problem is particularly awkward for text generation because prefixes are high-dimensional, non-Markov, and semantically heterogeneous. A growing body of work therefore revisits whether value heads are necessary at all in LLM post-training.

ReMax replaces learned critics with a greedy baseline and shows that much of PPO’s practical benefit can be retained with a simpler REINFORCE-style objective [12]. Back to Basics systematically compares PPO with RLOO-style estimators and finds that critic-free methods can match or exceed value-based baselines in RLHF [1]. REINFORCE++ further strengthens this line by improving robustness to prompt and reward-model variation without reintroducing a learned value function [9]. In reasoning-focused RLVR, GRPO normalizes rewards across a group of sampled responses rather than learning prefix values [19], and later work refines this family with theoretical analyses, off-policy variants, and sequence-level formulations such as GSPO [13, 30]. These methods substantially

86 simplify the optimization stack, but they generally still broadcast a trajectory-level signal across all
87 tokens once a scalar baseline is fixed.

88 2.3 Reasoning-Specific Analyses of RLVR

89 Another recent line asks why critic-free RLVR works so well for reasoning in the first place. A
90 Minimalist Approach to LLM Reasoning argues that strong reasoning gains can emerge from a
91 comparatively simple recipe that moves from rejection sampling to REINFORCE-style online updates,
92 suggesting that some of the field’s complexity is optional rather than essential [26]. Complementarily,
93 Wen et al. analyze RLVR theoretically and empirically, arguing that answer-level verifiable rewards
94 can nonetheless incentivize correct intermediate reasoning early in a trajectory [24]. These results
95 are important because they show that uniform outcome-level objectives already contain nontrivial
96 reasoning signal.

97 At the same time, they do not eliminate the question of *which* tokens should absorb that signal. Recent
98 empirical analysis of RLVR training dynamics suggests that improvements are disproportionately
99 driven by a minority of high-entropy “forking” tokens, and that restricting updates to this subset
100 can remain competitive or even outperform full-token updates in some settings [23]. This is the
101 closest recent line to our work conceptually: it supports the claim that token salience is highly
102 non-uniform during reasoning RL. Our work differs in scope and mechanism, however, by studying
103 general intrinsic weighting schemes based on token surprisal and entropy change rather than a hard
104 top-entropy masking rule or an empirical analysis specific to one RLVR recipe.

105 2.4 Fine-Grained Credit Assignment Beyond Uniform Token Updates

106 A large parallel literature attempts to make supervision denser than a single end-of-trajectory reward.
107 One family learns explicit token- or process-level reward estimators. Preference-grounded Token-level
108 Guidance derives token-level signals from preference data for fine-tuning [27], while Discriminative
109 Policy Optimization learns token-level Q-style reward models from preferences without requiring
110 fine-grained annotation [4]. PRIME pushes this direction further for reasoning tasks by constructing
111 implicit process rewards and updating process reward models online from rollouts and outcome
112 labels [6]. These methods can produce rich token-level feedback, but they rely on an auxiliary
113 reward-modeling component that our approach intentionally avoids.

114 Another family redistributes outcome rewards algorithmically rather than by training a dense reward
115 model. RED derives token-level rewards by redistributing holistic reward-model scores over a
116 trajectory [10]. SCAR uses Shapley-inspired attribution to estimate token or span contributions from
117 sequence-level feedback [2, 20]. TEMPO uses groups of sampled solutions to build a prefix tree and
118 compute nonparametric prefix values, yielding branch-sensitive corrections without a learned critic
119 [22]. GRPO- λ introduces eligibility traces and lambda-returns into critic-free RLVR to propagate
120 outcome information backward along the sequence [15]. Compared with these approaches, our focus
121 is narrower and cheaper: we do not train auxiliary reward models, estimate Shapley values, construct
122 trees, or generate corrective interventions, but instead ask how far one can get by reweighting the
123 existing policy-gradient terms using intrinsic distributional signals already present in the forward
124 pass.

125 2.5 Alternative Action Granularities and Token Selection

126 Several papers attack the same underlying problem by changing the optimization unit rather than
127 by redesigning the reward. MA-RLHF introduces macro actions, grouping tokens into higher-level
128 units to shorten the temporal distance between decisions and rewards [3]. GSPO moves importance
129 weighting and clipping from the token level to the sequence level for greater stability in large-scale
130 RL training [30]. VinePPO replaces learned value networks with Monte Carlo vine-style estimates
131 over continuations, seeking better credit propagation through structured rollout reuse [11]. These
132 methods reinforce the broader point that the granularity of optimization matters as much as the reward
133 definition.

134 Outside on-policy RL proper, token-selective optimization has also been explored in adjacent
135 paradigms. ConfPO emphasizes preference-critical tokens based on policy confidence in preference
136 optimization [28], and token weighting for long-range language modeling shows that non-uniform

token emphasis can improve optimization even in supervised settings [8]. Together with high-entropy token analyses in RLVR [23], these results strongly suggest that uniform token treatment is an arbitrary design choice rather than a necessity.

2.6 Positioning of This Work

The literature now makes three points clear. First, critics are not strictly required for effective LLM post-training: critic-free estimators such as RLOO, ReMax, GRPO, and GSPO are already competitive in both RLHF and RLVR [1, 12, 19, 30]. Second, many researchers have concluded that trajectory-level rewards are too coarse and have introduced denser supervision through process reward models, redistribution rules, tree structures, or temporal traces [10, 6, 22, 2, 15]. Third, recent reasoning-focused analyses indicate that only a subset of tokens may be responsible for most of the useful RL signal [24, 23].

Our paper sits at the intersection of these trends but occupies a distinct point in the design space. We study *intrinsic* token weighting for on-policy language RL: credit is redistributed at the token level using signals derived from the policy’s own predictive distribution, without learned value heads, external reward models, process annotations, tree construction, Shapley-style counterfactual attribution, or corrective rollouts. This positioning is important for reviewer clarity: our contribution is not another dense-reward model or another GRPO variant, but a minimal credit-redistribution mechanism that can be layered onto critic-free policy gradients while preserving their computational simplicity.

3 Methods

We consider on-policy reinforcement learning for autoregressive language models with trajectory-level rewards. Let π_θ denote a language model parameterized by θ , generating a completion $y = (y_1, \dots, y_T)$ conditioned on a prompt x . After sampling a full trajectory, the model receives a scalar reward $R(x, y) \in \mathbb{R}$ computed by an external verifier, such as exact-answer correctness or unit-test pass rate. Our objective is

$$J(\theta) = \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(\cdot|x)}[R(x, y)], \quad (1)$$

where $\mathcal{D}$ denotes the prompt distribution. We focus on the regime most relevant to recent reasoning-model training: sparse outcome rewards, on-policy sampling, and no learned value function.

3.1 Policy Gradient with Trajectory-Level Reward

For a fixed prompt x , the standard REINFORCE estimator is

$$g_{\text{rf}}(x, y) = R(x, y) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \quad (2)$$

In practice, a baseline $b(x)$ is subtracted to reduce variance without changing the expected gradient as long as $b(x)$ does not depend on the sampled action at token t :

$$g_{\text{base}}(x, y) = (R(x, y) - b(x)) \sum_{t=1}^T \nabla_\theta \log \pi_\theta(y_t \mid y_{<t}, x). \quad (3)$$

We use prompt-level multi-sample baselines. Given K sampled completions $\{y^{(i)}\}_{i=1}^K$ for the same prompt x with rewards $\{R^{(i)}\}_{i=1}^K$:

$$b_{\text{RLOO}}^{(i)}(x) = \frac{1}{K-1} \sum_{j \neq i} R^{(j)} \quad (4)$$

is the leave-one-out baseline used in RLOO-style estimators, while GRPO-style normalization can be written as

$$A_{\text{GRPO}}^{(i)}(x) = \frac{R^{(i)} - \mu_R(x)}{\sigma_R(x) + \varepsilon}, \quad (5)$$

where $\mu_R(x)$ and $\sigma_R(x)$ are the within-prompt mean and standard deviation of the K rewards. In both cases, the resulting scalar advantage is then broadcast uniformly over all tokens in the sampled trajectory. This scalar-broadcast assumption is precisely what we relax.

175 3.2 Token-Level Credit Redistribution

176 We introduce token-level weights $w_t(x, y)$ that redistribute a trajectory-level advantage across tokens:

$$g_w(x, y) = A(x, y) \sum_{t=1}^T w_t(x, y) \nabla_{\theta} \log \pi_{\theta}(y_t \mid y_{<t}, x), \quad (6)$$

177 where $A(x, y)$ denotes either $R(x, y) - b_{\text{RLOO}}(x)$ or $A_{\text{GRPO}}(x, y)$ depending on the baseline choice.

178 The weights satisfy

$$\sum_{t=1}^T w_t(x, y) = 1, \quad w_t(x, y) \geq 0. \quad (7)$$

179 This normalization keeps the total update magnitude comparable across weighting schemes and
180 isolates the effect of credit redistribution from simple rescaling.

181 Uniform token credit corresponds to $w_t = 1/T$ for all t . Our goal is to replace this uniform allocation
182 with *intrinsic* weights derived from quantities already produced by the policy during generation. We
183 do not train token-level reward models, estimate prefix values, or build explicit search trees.

184 3.3 Intrinsic Token-Weighting Mechanisms

185 We define each method through an unnormalized salience score $\alpha_t(x, y) \geq 0$ and then normalize
186 within each trajectory:

$$w_t(x, y) = \frac{\alpha_t(x, y) + \varepsilon}{\sum_{k=1}^T (\alpha_k(x, y) + \varepsilon)}, \quad (8)$$

187 with a small $\varepsilon > 0$ to avoid degenerate all-zero cases. In implementation, the weights are treated
188 as *detached* scalars when multiplying the policy-gradient term, so the update does not introduce
189 second-order derivatives through the weighting function itself. This keeps the optimization rule close
190 in spirit and cost to standard critic-free policy gradients.

Uniform Weighting (Baseline).

$$w_t^{\text{uniform}} = \frac{1}{T}. \quad (9)$$

191 Combined with an RLOO or GRPO baseline, this recovers the standard critic-free estimator that
192 assigns identical credit to every token in the sampled completion.

193 **Surprisal Weighting.** Our first intrinsic score is token surprisal,

$$\alpha_t^{\text{sup}}(x, y) = -\log \pi_{\theta}(y_t \mid y_{<t}, x). \quad (10)$$

194 This emphasizes low-probability decisions under the current policy. Intuitively, these are tokens
195 where the model commits to a less routine continuation and where uniform credit assignment may be
196 most diluted by predictable filler tokens.

197 **Entropy-Reduction Weighting.** Our second intrinsic score measures local entropy reduction. Let

$$H_t^{\text{pre}}(x, y) = -\sum_v \pi_{\theta}(v \mid y_{<t}, x) \log \pi_{\theta}(v \mid y_{<t}, x) \quad (11)$$

198 denote the predictive entropy before sampling token y_t , and let

$$\Delta H_t^{\text{ent}}(x, y) = \max(0, H_t^{\text{pre}}(x, y) - H_{t+1}^{\text{pre}}(x, y)) \quad (12)$$

199 for $t < T$. For the final token we set $\Delta H_T^{\text{ent}} = 0$. We then use

$$\alpha_t^{\text{ent}}(x, y) = \Delta H_t^{\text{ent}}(x, y). \quad (13)$$

200 This score highlights tokens after which the model’s next-step distribution becomes substantially
201 more concentrated. Such events often correspond to commitment points in reasoning, where one
202 branch of continuation becomes much more likely than its alternatives.

Length-Robust Normalization. Because reasoning trajectories can vary substantially in length, all weights are normalized within each sampled completion rather than across the minibatch. This ensures that longer trajectories do not automatically receive larger aggregate updates simply because they contain more token positions with nonzero salience. The comparison between weighting schemes therefore reflects *where* credit is assigned within a trajectory, not how much total credit a longer sequence receives.

Batch-Level Baselines. All weighting schemes can be paired with either RLOO or GRPO-style prompt-level baselines. Unless otherwise specified, we use the same rollout groups, optimizer, sampling hyperparameters, and baseline family across methods. This isolates token redistribution as the only algorithmic difference.

3.4 Relationship to Existing Methods

Our formulation recovers standard critic-free RL as a special case: uniform weighting plus a prompt-level batch baseline yields the usual RLOO/GRPO-style estimator. It also clarifies how our method differs from nearby alternatives. Unlike critic-based PPO, we do not learn a value head. Unlike RED, PRIME, or token-level reward-model methods, we do not construct dense rewards from an auxiliary model. Unlike TEMPO or related tree-based methods, we do not build prefix graphs or estimate branch values. Unlike hard token-selection methods based on top-entropy masking, we retain a dense token update but modulate it continuously using intrinsic salience scores.

The resulting estimator is intentionally minimal. It changes only the within-trajectory allocation of a scalar on-policy advantage, using quantities already available from the forward pass. This makes it easy to layer on top of existing critic-free RLVR pipelines while preserving their computational simplicity.

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